

AI Training Program

Comprehensive Lab Review & Cheat Sheet

1. Curriculum Overview

This document summarizes the 8-lab journey from data hygiene to predictive artificial intelligence.

Lab	Topic	Key Objective	Core Algorithm/Library
02	EDA & Data Cleaning	Identify best advertising channel	Pandas, Matplotlib (Correlation)
03	NumPy & Pandas Practice	Salary growth analysis	NumPy (np.where), Logic
04	Predictive Regression	House price estimation	Scikit-Learn (LinearRegression)
05	Classification Lab	Diabetes screening	LogisticRegression, Confusion Matrix
06	Neural Network Basics	Handwritten digit recognition	MLPClassifier (Deep Learning)
07	Prompt Engineering	AI Instruction Framework	Role, Task, Context, Constraints
08	Capstone Project	Student Performance Prediction	Full ML Lifecycle

2. Data Science Workflow "Formula"

Every professional AI project follows this sequence:

1. **Ingestion:** Load the raw samples (`pd.read_csv()`).
2. **Cleaning:** Remove errors and duplicates (`isnull()`, `drop()`).
3. **EDA:** Find the story using visuals and correlation (`corr()`, `plt.scatter()`).
4. **Modeling:** Split data and train the "brain" (`train_test_split`, `model.fit()`).
5. **Evaluation:** Check the accuracy (MAE, R2, Confusion Matrix).

3. Essential Code Cheat Sheet

A. Data Cleaning & Inspection

```
df.info()                # Check data types
df.isnull().sum()        # Find missing values
df.describe().T          # Statistical summary
df.drop_duplicates()      # Remove identical records
```

B. Feature Engineering (NumPy)

```
# Creating labels based on logic
df['Status'] = np.where(df['Score'] >= 75, 'Pass', 'Fail')

# Converting Text to Numbers
df = pd.get_dummies(df, drop_first=True)
```

C. Machine Learning (Regression & Classification)

```
# Splitting for Training and Testing
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# Training a Model
model.fit(X_train, y_train)

# Accuracy Metrics
print(accuracy_score(y_test, predictions))
print(mean_absolute_error(y_test, predictions))
```

4. Capstone Findings (Lab 08)

In the student performance analysis, we identified that study habits and attendance are the primary drivers of academic success.

- **Strongest Predictor:** Study Hours (highest correlation).
- **Model Reliability:** The model provides a baseline for early intervention.
- **Ethical Note:** AI should be used as a support tool, not a final decision-maker.

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